# Mathematical Background: Topology Generator from a Subbasis

## Scope

$X$ is a fixed finite set throughout. This document covers: generating a topology from a subbasis (`generator.py`); topology validation (`validate.py`); separation axioms $T_0$–$T_4$ (`separation.py`); connectedness (`connectedness.py`); compactness (`compactness.py`); and continuity (`continuity.py`).

## 1 Topologies, Bases, Subbases

**Definition 1.** *A* topology *on $X$ is $\mathcal{T} \subseteq \mathcal{P}(X)$ with $\emptyset, X \in \mathcal{T}$, closed under arbitrary unions and finite intersections.*

**Definition 2.** *A* basis $\mathcal{B}$ *generates* $\mathcal{T} = \{\bigcup \mathcal{U} : \mathcal{U} \subseteq \mathcal{B}\}$. *A* subbasis $\mathcal{S}$ *generates the basis* $\mathcal{B} = \{\bigcap \mathcal{F} : \mathcal{F} \subseteq \mathcal{S},\ |\mathcal{F}| < \infty\} \cup \{X\}$ *(the empty intersection is $X$), and hence $\mathcal{T}$.*

Since $X$ is finite, $\mathcal{P}(X)$ is finite, so "arbitrary union" and "finite union" coincide, and every finite intersection is reachable by intersecting one subbasis element at a time. `generate_basis_from_subbasis` computes $\mathcal{B}$ by closing $\{X\}$ under pairwise intersection with each $s \in \mathcal{S}$ to a fixed point; `generate_topology_from_b` closes $\{\emptyset\} \cup \mathcal{B}$ under pairwise union with each $b \in \mathcal{B}$ the same way. Both terminate since $\mathcal{P}(X)$ is finite and each step either adds a new set or reaches the fixed point.

**Proposition 1.** *For finite $X$, checking $\mathcal{T}$ closed under* pairwise *union/intersection of its own members is equivalent to closure under arbitrary unions and finite intersections.*

*Proof.* Any union (intersection) of a subset of $\mathcal{T}$ is built by repeatedly combining two elements at a time; if $\mathcal{T}$ is closed under each such step, by induction on $|\mathcal{T}| < \infty$ it is closed under the full operation. □

This is what `is_topology` checks directly, with a counterexample pair returned on failure.

## 2 Separation Axioms

For distinct $x, y \in X$:

| | |
|---|---|
| $T_0$ | some open set contains exactly one of $x, y$ |
| $T_1$ | some open set contains $x$ not $y$, and some open set contains $y$ not $x$ |
| $T_2$ | disjoint open sets separate $x$ and $y$ |

**Proposition 2.** $T_1 \iff$ *every singleton is closed.*

*Proof.* ($\Rightarrow$) For fixed $x$, $X \setminus \{x\} = \bigcup_{y \neq x} U_y$ where $U_y$ is open, contains $y$, excludes $x$ (exists by $T_1$); a union of opens is open, so $X \setminus \{x\}$ is open, i.e. $\{x\}$ is closed. ($\Leftarrow$) If every singleton is closed, $X \setminus \{y\}$ is an open set containing $x$ but not $y$, and symmetrically for $X \setminus \{x\}$. □

`check_T1_via_closed_singletons` implements the right-hand side independently, as a cross-check on `check_T1`.

**Definition 3** (Munkres)**.** *$X$ is* regular *if for every closed $C$ and $x \notin C$, disjoint open sets separate $\{x\}$ and $C$;* normal *if every pair of disjoint closed sets is so separated. $T_3 = \text{regular} \wedge T_1$; $T_4 = \text{normal} \wedge T_1$.*

**Theorem 1.** *Every finite $T_1$ space is discrete.*

*Proof.* By the Proposition above, $T_1$ makes every singleton closed. $X$ finite $\Rightarrow$ every subset $A \subseteq X$ is a finite union of singletons, $A = \bigcup_{a \in A} \{a\}$; each $X \setminus \{a\}$ is open (singletons closed), and $A = \bigcap_{a \notin A} (X \setminus \{a\})$ is a finite intersection of opens, hence open. So every subset is open: $\mathcal{T} = \mathcal{P}(X)$. □

**Corollary 1.** *On a finite space, $T_1 \iff T_2 \iff T_3 \iff T_4 \iff$ discrete.*

*Proof.* $T_2 \Rightarrow T_1$ always (fewer conditions). $T_1 \Rightarrow$ discrete (Theorem 1). Discrete $\Rightarrow T_2$ (singletons separate any two points) $\Rightarrow$ discrete is regular and normal trivially (any two disjoint sets are already separated by themselves, both open), so discrete $\wedge\, T_1 \Rightarrow T_3, T_4$. $T_3, T_4 \Rightarrow T_1$ by definition. All five are therefore equivalent. □

Checked computationally across 500 random finite topologies in `test_t1_t2_t3_t4_and_discrete_all_coincide` for every generated topology, `is_T1 == is_T2 == is_T3 == is_T4 == is_discrete`.

## 3 Connectedness

**Definition 4.** *$A \subseteq X$ is* clopen *if $A$ and $X \setminus A$ are both open. $X$ is* connected *if its only clopen subsets are $\emptyset, X$.*

**Proposition 3.** *The clopen subsets of a finite space form a Boolean algebra under $\cap, \cup, ^c$, and for $x \in X$, $\text{comp}(x) := \bigcap \{C \text{ clopen} : x \in C\}$ is the connected component of $x$.*

*Proof sketch.* Closure of clopen sets under $\cap, \cup, ^c$ is immediate from the definition of a topology. $\text{comp}(x)$ is a finite intersection of clopen sets, hence clopen, and by minimality no proper nonempty clopen subset of $\text{comp}(x)$ can separate it further; distinct $\text{comp}(x)$'s are disjoint or equal since their intersection, if nonempty, would itself be a smaller clopen set containing both would-be centers, contradicting minimality unless they coincide. □

`connected_components` computes $\text{comp}(x)$ for each $x$ directly from this characterization. The discrete topology: every singleton is clopen, so each $\text{comp}(x) = \{x\}$ ($n$ components). The indiscrete topology: only clopen sets are $\emptyset, X$, so one component. The Sierpiński space ($\{0, 1\}, \{\emptyset, \{0\}, \{0, 1\}\}$): $\{0\}$ is open but $\{1\}$ (its complement) is not, so $\{0\}$ is not clopen; the only clopen sets are $\emptyset, X$, so it is connected despite not being $T_1$.

## 4 Compactness

**Definition 5.** *$X$ is* compact *if every open cover has a finite subcover.*

**Proposition 4.** *Every finite topological space is compact.*

*Proof.* $\mathcal{T} \subseteq \mathcal{P}(X)$ and $X$ finite $\Rightarrow |\mathcal{T}| \leq 2^{|X|} < \infty$. Any open cover is a subset of $\mathcal{T}$, hence itself finite, hence trivially its own finite subcover. □

`is_compact` verifies this by constructing an explicit finite subcover via greedy set cover (`has_finite_subcover`) rather than returning a hardcoded constant — the content here is the proof that this is always possible, not the search itself.

## 5 Continuity

**Definition 6.** *$f : (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$ is* continuous *if $f^{-1}(V) \in \mathcal{T}_X$ for every $V \in \mathcal{T}_Y$. A* homeomorphism *is a continuous bijection with continuous inverse.*

**Proposition 5.** *If $\mathcal{T}_X$ is finer than $\mathcal{T}'_X$ (i.e. $\mathcal{T}'_X \subseteq \mathcal{T}_X$), then* $\mathrm{id} : (X, \mathcal{T}_X) \to (X, \mathcal{T}'_X)$ *is continuous.*

*Proof.* $\mathrm{id}^{-1}(V) = V$ for every $V \in \mathcal{T}'_X \subseteq \mathcal{T}_X$. □

In particular id : discrete → any topology is always continuous (discrete is finest), and id : any topology → discrete is continuous only if the source is already discrete — confirmed directly: `indiscrete→discrete` fails, with a witnessing non-open preimage returned by `is_continuous`. A constant map is continuous regardless of $\mathcal{T}_X$, since $f^{-1}(V) \in \{\emptyset, X\}$ for every $V$, and both are always open. `is_homeomorphism` checks $f, f^{-1}$ are genuine mutual-inverse bijections before checking continuity in both directions.